Unit: 6 Unit 8: Quadratics
Topic: Vertex Form and Intercept Form

Name: ____________
Date: ________

Quiz (Carolina Reaper)

Q1. Write the quadratic function $f(x) = x^2 + 4x + 4$ in vertex form.
(A) $f(x) = (x + 2)^2$
(B) $f(x) = (x - 2)^2$
(C) $f(x) = x^2 + 4x + 4$
(D) $f(x) = (x + 2)^2 - 4$
(E) $f(x) = (x - 2)^2 + 4$

Q2. Which of the following is the vertex form of a quadratic function?
(A) $f(x) = ax^2 + bx + c$
(B) $f(x) = a(x - h)^2 + k$
(C) $f(x) = x^2 + bx + c$
(D) $f(x) = ax^2 - bx - c$
(E) $f(x) = (x - a)^2 + b$

Q3. Write the quadratic function $f(x) = (x-2)^2+1$ in standard form.
(A) $f(x) = x^2 - 4x + 3$
(B) $f(x) = x^2 - 4x + 5$
(C) $f(x) = x^2 + 4x + 5$
(D) $f(x) = x^2 - 4x + 4$
(E) $f(x) = x^2 + 4x + 3$

Q4. What is the vertex of the quadratic function $f(x) = (x + 1)^2 - 2$?
(A) $(-1, 2)$
(B) $(-1, -2)$
(C) $(1, -2)$
(D) $(1, 2)$
(E) $(0, 0)$

Q5. Which of the following is the intercept form of a quadratic function?
(A) $f(x) = a(x - p)(x - q)$
(B) $f(x) = ax^2 + bx + c$
(C) $f(x) = (x - p)^2 + q$
(D) $f(x) = a(x + p)(x + q)$
(E) $f(x) = x^2 + bx + c$

Q6. Write the quadratic function $f(x) = 2(x - 1)(x + 2)$ in standard form.
(A) $f(x) = 2x^2 + 2x - 4$
(B) $f(x) = 2x^2 - 2x - 4$
(C) $f(x) = 2x^2 + 2x + 4$
(D) $f(x) = 2x^2 - 2x + 4$
(E) $f(x) = x^2 + 2x - 4$

Q7. What are the x-intercepts of the quadratic function $f(x) = (x + 2)(x - 3)$?
(A) $(-2, 0)$ and $(3, 0)$
(B) $(-2, 0)$ and $(-3, 0)$
(C) $(2, 0)$ and $(3, 0)$
(D) $(2, 0)$ and $(-3, 0)$
(E) $(-2, 0)$ and $(0, 0)$

Q8. Write the quadratic function $f(x) = x^2-5x-6$ in intercept form.
(A) $f(x) = (x - 6)(x + 1)$
(B) $f(x) = (x + 6)(x - 1)$
(C) $f(x) = (x + 2)(x - 3)$
(D) $f(x) = (x - 2)(x + 3)$
(E) $f(x) = (x + 1)(x - 6)$

Open Ended Questions (FRQ)

Q9. Write the quadratic function $f(x) = 3x^2 - 12x - 4$ in vertex form and find its vertex. Show your work and explain your reasoning.

Q10. Write the quadratic function $f(x) = (x - 2)(x + 4)$ in standard form and find its x-intercepts. Show your work and explain your reasoning.

Chef's Tips

- Check that students show their work and explain their reasoning for free-response questions.
- Consider allowing students to use graphing calculators to check their answers.
- Provide feedback to students on their understanding of vertex and intercept forms.